

## Project Initialization and Planning Phase

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 5 July 2026                                                 |
| Team ID       | xxx                                                         |
| Project Title | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                     |

### Project Proposal (Proposed Solution) report

| Project Overview  |                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | Develop an AI-based crop recommendation system that predicts the most suitable crop using soil nutrients and climate conditions.                           |
| Scope             | The system analyzes agricultural data, trains machine learning models, and provides accurate crop recommendations through a web application.               |
| Problem Statement |                                                                                                                                                            |
| Description       | Farmers often face difficulties in selecting suitable crops due to changing soil and climate conditions, leading to low productivity and financial losses. |
| Impact            | The system improves crop selection, increases agricultural productivity, reduces farming risks, and supports sustainable farming.                          |

## Proposed Solution

|              |                                                                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approach     | Use machine learning algorithms (KNN, Logistic Regression, Decision Tree, Random Forest) to analyze soil and climate data and recommend the best crop.                                                                                                  |
| Key Features | <ul style="list-style-type: none"> <li>• AI-based crop recommendation</li> <li>• Soil and climate analysis</li> <li>• Fast and accurate predictions</li> <li>• User-friendly web interface</li> <li>• Prediction reports and recommendations</li> </ul> |

## Resource Requirements

| Resource Type       | Description                           | Specification/Allocation |
|---------------------|---------------------------------------|--------------------------|
| <b>Hardware</b>     |                                       |                          |
| Computing Resources | CPU                                   | Intel Core i5 or higher  |
| Memory              | RAM specifications                    | 8 GB                     |
| Storage             | Disk space for data, models, and logs | 1 TB SSD                 |
| <b>Software</b>     |                                       |                          |
| Frameworks          | Python frameworks                     | Flask                    |

|                         |                      |                                                  |
|-------------------------|----------------------|--------------------------------------------------|
| Libraries               | Additional libraries | scikit-learn, pandas, numpy, matplotlib, seaborn |
| Development Environment | IDE                  | Jupyter Notebook, pycharm                        |
| Data                    | Source, size, format | Kaggle dataset, 614, csv UCI dataset, 690, csv   |